

9506

4. (a) Give an account on the role of hormones in maintenance of pregnancy. 8
(b) Explain the formation and functions of corpus luteum. 7
5. (a) Give an account of sexual differentiation and developmental abnormalities in males. 8
(b) Describe hormonal regulation of parturition. 7
6. (a) Elaborate the changes in mammary glands during lactation. Add a note on its hormonal regulation. 8
(b) Analyze the role of the placenta as a multi-organ system for the developing fetus. 7
7. Write short notes on any **three** of the following :
5+5+5=15
(a) Biomarkers of Endometrial Cancer
(b) Androgen insensitivity syndrome
(c) Dysregulation of spermatogenesis
(d) Labor abnormalities.

[This question paper contains 4 printed pages]

Your Roll No. : 14.06.2026 (E)

Sl. No. of Q. Paper : 9506 I

Unique Paper Code : 223352004

Name of the Paper : DSE-23 Reproductive Endocrinology

Name of the Course : B.Sc.(H) Zoology/B.Sc. (P) Life Sciences

Semester : VIII (NEP-UGCF)

Time : 3 Hours **Maximum Marks : 90**

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
(b) Attempt any **five** questions in **all**, including Question No. 1 which is compulsory.
(c) **All** questions carry equal marks.
(d) Draw well labelled diagram wherever necessary.

1. (a) Define 6
(i) Hyperprolactinemia
(ii) Amenorrhea

- (iii) Follicular Atresia
 - (iv) Cryptorchidism
 - (v) Menopause
 - (vi) Pre-eclampsia
- (b) State whether the statement is true or false : 6
- (i) The pulsatile secretion of GnRH is crucial for a normal menstrual cycle.
 - (ii) Endocrine disruptors like BPA cannot cross placenta.
 - (iii) Progesterone is dominant during the luteal phase.
 - (iv) Inhibin B is secreted by Sertoli cells.
 - (v) Prenatal exposure to endocrine disruptors cannot affect testicular development.
 - (vi) HER2-negative cancers cannot be treated.
- (c) Expand the following : 6
- (i) IUGR
 - (ii) FSH
 - (iii) ELISA
 - (iv) hCG
 - (v) PCOS
 - (vi) EDCs

- (d) Differentiate between the following pairs : 8
- (i) Inhibin and Activin
 - (ii) Spermatogenesis and Oogenesis
 - (iii) Bulbourethral gland and Prostate gland
 - (iv) Secondary Follicle and Graffian Follicle
- (e) Fill in the blanks : 4
- (i) Neuroendocrine regulation involves interaction between, pituitary, and gonads.
 - (ii) High estrogen levels cause feedback leading to LH surge.
 - (iii) The follicle transforms into after ovulation.
 - (iv) Kisspeptin neurons regulate secretion.
2. (a) Explain the neuroendocrine control of testicular functions. 8
- (b) Discuss the accessory reproductive glands in males and their role in reproduction. 7
3. (a) Give a brief account on dysregulation of menstrual cycle. 8
- (b) Explain the physiology of ovary and the role of ovarian hormones in female reproductive system. 7